

Abhinandan

+91-8130635690 | abhinandan@abhinandan.one

abhinandan.one | linkedin.com/in/abhibuilds | github.com/awesome-pro

EXPERIENCE

Founding Software Engineer

Sept 2025 – Present

Browser

Remote

- Built Browser's production Chrome MV3 recorder + CDP-native browser agent, achieving 95%+ precise AX/DOM element capture with iframe support, obstruction checks, and real mouse/key/upload execution.
- Built a smart streaming ReAct loop across FastAPI + extension with SSE tool execution, multi-tab orchestration, safe parallelism, abort/continue, and audit logs.
- Cut automation LLM spend by roughly 67% using compact recording traces, context-window compression, prompt caching, and model-routing across GPT-5, Claude Sonnet 4.6, and Gemini 3.1 Pro.
- Shipped a zero-LLM replay engine: recordings run as variable-driven tool-call templates, with a stateful AI fallback that resumes mid-run on failure instead of restarting.
- Shipped self-healing docs that auto-repair on UI drift — Haiku → Sonnet diff triage, LLM-free replay of intact steps, and a CDP agent that fixes only what changed.

Software Engineer

Jan 2025 – Jul 2025

Cynos Nexus

Noida, India

- Shipped core features of an AI-powered real estate platform using Next.js, Nest.js, GraphQL, Redis, and GCP.
- Built the AI knowledge base service on FastAPI + LangChain.
- Integrated Google Cloud Vision + XLSX import to let users seed client databases from contact diary photos or Excel.
- Automated deployment infrastructure using Docker, GitHub Actions for CI/CD, and Nginx for reverse proxy/load balancing.

Contract Software Engineer

Sept 2024 – Dec 2024

Etkin.ai

Remote, Turkiye

- Engineered a high-performance PDF generator using WeasyPrint, converting dynamic JSON reports to high-quality PDFs.
- Optimised data processing using CSV upload & Celery worker, reducing processing time by 40%.
- Established CI/CD pipelines with GitHub Actions and Docker on AWS EC2.
- Collaborated to build a LangChain and pgvector knowledge base, improving ML query accuracy by 15%.

Open Source Contributor

June 2024 – Aug 2024

HeroUI

Y Combinator S24 startup

- Received a **personal offer from the CEO** to join HeroUI (prev. NextUI) after making **open-source contributions**.
- Resolved 10+ bugs & delivered 7+ feature enhancements in core components including Calendar, Table and Pagination.

SWE Intern

Jan 2024 – Apr 2024

SkilledUp

Noida

OPEN SOURCE PROJECTS

AgentFlow-Pro (*Research*) | *PyTorch, TRL, DAPO, PRM, PEFT/LoRA, Ollama, FastMCP* | github.com/awesome-pro/agentflow-pro

- Rebuilt and extended the ICLR 2026 *AgentFlow* architecture as a local Qwen3-8B Planner → Executor → Verifier → Memory loop (grammar-constrained planning, Tavily search, sandboxed Python/SymPy tools); trained the planner with **DAPO + a learned Qwen3-0.6B Process Reward Model** (replacing the paper's GRPO) and a hand-built dynamic-sampling stage. TRL lacks, then ran the full pipeline on an A40 (DeepSeek-judge step labels → PRM → bf16 LoRA DAPO → GGUF) and measured **+5.0 pts on GPQA-Diamond (40.0%→45.0%)** under leakage-free, quantization-matched evaluation.

guardloop | *OpenAI/Anthropic SDKs, OpenTelemetry, LangGraph* | github.com/awesome-pro/guardloop

- Production guardrail runtime for async agents: pre-flight Decimal cost/token/time/tool-call budgets, per-tool circuit breakers, verifier feedback retry loop, OpenTelemetry GenAI spans, and no-rewrite LangGraph/OpenAI Agents SDK adapters that enforce safety inside existing LLM/tool calls.

smartmemo | *PyTorch, sentence-transformers, FAISS, SQLite* | github.com/awesome-pro/smartmemo

- Semantic memory/cache for LLM agents where embeddings retrieve candidates and a learned pair-equivalence classifier decides reuse; shipped bundled classifier-v2 trained on 16,576 pairs across 9 domains (+30 precision points at equal recall vs cosine), plus feedback export, implicit bad-hit detection, gated retraining, and WAL-backed SQLite hardening.

orchflow | *Asyncio, LiteLLM* | github.com/awesome-pro/orchflow

- Dependency-free typed Python framework for readable multi-agent pipelines: sequential/parallel/conditional flows, retries, shared **StepContext**, flat traces, lifecycle events, human gates, JSON checkpoint/resume, and optional LiteLLM Agent with structured JSON/Pydantic outputs.

agenteval | *AsyncIO, OpenAI/Anthropic/LangChain adapters* | github.com/awesome-pro/agenteval

- Agent evaluation toolkit replacing exact-match asserts with repeated-run pass-rate tests; traces tool calls/timing/steps, supports collect-then-raise behavioral assertions, OpenAI/Anthropic/LangChain adapters, and Typer CLI JSON reports for CI gates.

TECHNICAL SKILLS

ML & Retrieval: PyTorch, Hugging Face, sentence-transformers, embeddings, classifier training, RAG, FAISS, Pinecone, Qdrant, Weaviate, ChromaDB, pgvector
AI Agents & Frameworks: LangGraph, OpenAI Agents SDK, LangChain, LlamaIndex, CrewAI, AutoGen, Pydantic AI, LiteLLM, MCP / FastMCP, multi-agent orchestration, ReAct, function calling, tool use, CDP-based browser agents
LLM Engineering: Fine-tuning (LoRA, QLoRA, PEFT), RLHF / DPO / GRPO / DAPO / PRM, quantization (AWQ, GPTQ, GGUF), inference engines (vLLM, SGLang, Ollama, TGI), context engineering, KV-cache strategies, streaming
Observability & Eval: OpenTelemetry, Langfuse, Arize Phoenix, LangSmith, statistical pass-rate evaluation, AIME / GPQA benchmarks
Cloud & Infra: Docker, GCP (Vertex AI), AWS, Nginx, Redis, PostgreSQL, MongoDB
Backend: FastAPI, Nest.js, Node.js, GraphQL, WebSockets, REST, SSE
Languages: Python, TypeScript, C++, SQL, Bash
Frontend: Next.js, React, Redux, Turbopack, Edge Runtime

AWARDS & RECOGNITION

International Youth Math Challenge Gold Honour | **Top 1% TypeScript Engineer Globally** (Algora) | **Amazon ML Summer School 2025** | **Reliance Foundation Scholar** | **HDFC Badhate Kadam Scholar** | 2nd Place, Outlier AI Hackathon |

EDUCATION

Dr. APJ Abdul Kalam Technical University <i>B.Tech, Computer Science</i>	Nov 2022 – Apr 2026 <i>India</i>
--	-------------------------------------